Loops and Data Collections Notes

While Loop	

Python

```
def rocket_launch_countdown():
    seconds_left = 10

    while seconds_left > 0:
        print(f"T-minus {seconds_left}...")
        seconds_left -= 1

    print("Liftoff!")
    print()

rocket_launch_countdown()
```

Python

```
def atm_withdrawal_simulator():
    balance = 500
    withdrawal_amount = 120

    while balance >= withdrawal_amount:
        balance -= withdrawal_amount
        print(f"Withdrew ${withdrawal_amount}. Remaining: ${balance}")

    print(f"Cannot withdraw ${withdrawal_amount} anymore. Final: ${balance}")
    print()
atm_withdrawl_simulator
```

For Loop	

Python

```
def workout_rep_counter():
    total_reps = 8

    for rep in range(total_reps):
        print(f"Push-up #{rep + 1}")
    print()
workout_rep_counter()
```

List	

Python

```
def grocery_list_manager():
    groceries = ["Milk", "Eggs", "Bread"]

    groceries.append("Apples")
    groceries.append("Coffee")
    print(f"After shopping trip add-ons: {groceries}")

    groceries[0] = "Almond Milk"
    print(f"Swapped first item: {groceries}")

    removed_item = groceries.pop(2)
    print(f"Removed '{removed_item}' from the list: {groceries}")

    # Check how many items are on the list
    print(f"Total items to buy: {len(groceries)}")
    print(f"Last item on the list: {groceries[-1]}")
```

Index	

Python

```
def credit_card_masker():
    card_number = "4471502938467123"

    last_four = card_number[-4:]
    print(f"Full card number: {card_number}")
    print(f"Last four digits: {last_four}")

    hidden_portion = card_number[:-4]
    print(f"Digits to hide: {hidden_portion}")

    masked_number = ("*" * len(hidden_portion)) + last_four
    print(f"Masked number for the receipt: {masked_number}")

    network_digit = card_number[0]
    print(f"First digit (used to identify card network): {network_digit}")
    print()
credit_card_masker()
```

Python

```
def playlist_player():
    playlist = ["Bohemian Rhapsody", "Hotel California", "Imagine"]

    for song in playlist:
        print(f"Now playing: {song}")
    print()

playlist_player()
```

Python

```
def classroom_seating_chart():
    students = ["Maria", "James", "Priya", "Diego", "Sam"]

    for i in range(len(students)):
        seat_number = i + 1
        print(f"Seat {seat_number}: {students[i]}")
    print()

classroom_seating_chart()
```

Python

```
def product_lookup():
    product = {
        "name": "Wireless Headphones",
        "price": 79.99,
        "in_stock": True,
        "sku": "WH-2024-BLK"
    }

    print(f"Product: {product['name']}")
    print(f"Price: ${product['price']}")

    key_to_check = "discount"
    if key_to_check in product:
        print(f"Discount available: {product[key_to_check]}")
    else:
        print(f"No '{key_to_check}' key found on this product yet.")

    product["discount"] = 0.15
    print(f"Added discount info: {product}")

    product.pop("sku")
    print(f"After removing SKU: {product}")
    print()
```

Python

```
def weather_station_dashboard():
    forecast = [
        {"day": "Monday", "high_temp": 72, "condition": "Sunny"},
        {"day": "Tuesday", "high_temp": 65, "condition": "Cloudy"},
        {"day": "Wednesday", "high_temp": 58, "condition": "Rainy"},
    ]

    for data in forecast:
        print(f"{data['day']}: {data['high_temp']}°F, {data['condition']}")

    index = 0
    found_rain = False
    while index < len(forecast) and not found_rain:
        if forecast[index]["condition"] == "Rainy":
            print(f"First rainy day this week: {forecast[index]['day']}")
            found_rain = True
        index += 1
    print()
```